

Vivek Sai Manthri

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EDUCATION

Rutgers University

- M.S. in Computer Science | GPA: 3.92/4.00
- B.S. in Computer Science, Data Science | GPA: 3.84/4.00

New Brunswick, NJ
Sep. 2025 – May 2027
Sep. 2022 – May 2025

EXPERIENCE

Regeneron

Warren, NJ

Clinical Study Start-Up Automation Intern

Jun. 2026 – Aug. 2026

- Developed production-ready enterprise AI agents in Microsoft Copilot Studio using Power Automate Agent Flows and M365 integrations to automate Clinical Study/Site Start-Up (SSU) processes and eliminate repetitive manual work
- Engineered SSU Data Steward FilePilot, an end-to-end automation solution that processes Jira ticket data, updates Excel workbooks using Python, JavaScript, and TypeScript scripts, and publishes files to SharePoint, saving 5+ hours daily.
- Built AI-powered site/clinical operations solutions reducing site intelligence report preparation time by 50% and developed PQ DraftPilot, a multi-agent workflow accelerating Performance Qualification Script generation by 75%

Gogentic AI

Houston, TX

Full-Stack Developer Intern

Jul. 2025 – Aug. 2025

- Developed NeuroVault, an AI memory system, using React, TypeScript, Tailwind CSS, and Vite, reducing load times by 30% and increasing user engagement
- Engineered FastAPI/Python REST APIs for audio/PDF processing with SQLite and Neo4j local-first storage, reducing cloud costs while achieving sub-200ms latency
- Implemented RAG pipelines using ChromaDB, OpenAI embeddings, and LangChain for semantic search across 10K+ documents, improving query accuracy by 25%; containerized deployment with Docker, accelerating releases by 40%

Aroopa, Inc

Monroe Township, NJ

Software Developer Intern

Jun. 2025 – Aug. 2025

- Built a production-ready MERN stack application using React, Node.js, and Express, scaling to support 8K+ monthly users
- Created responsive UI components with React, Tailwind CSS, and HTML, increasing user engagement by 15% and improving the overall user experience
- Architected MongoDB schemas and REST APIs, optimizing queries for 40% faster data retrieval; performed unit and integration testing to resolve 10+ functional issues and improve reliability

Memorial Sloan Kettering Cancer Center

New York, NY

Software Engineering Intern

Jun. 2024 – Aug. 2024

- Engineered a Conditional GAN to generate privacy-preserving synthetic clinical data, reducing manual data creation by 30% while maintaining high fidelity to real patient data
- Optimized ML and data preprocessing pipelines using TensorFlow, Keras/PyTorch, OneHotEncoder, Scikit-learn, and Pandas, improving computational efficiency by 25% and streamlining ETL workflows
- Containerized the ML environment with Docker and integrated it into a CI/CD pipeline, improving deployment speed and reproducibility while reducing configuration discrepancies

PROJECTS

Online Auction System | JavaScript, Java, MySQL, JSP, JDBC, HTML/CSS, AWS (EC2, RDS), Apache Tomcat

- Designed a secure web-based, full-stack auction platform using Eclipse, featuring user authentication, real-time bidding, and automated alert notifications, deployed on AWS cloud infrastructure (EC2 + MySQL RDS database), reducing latency by 40%
- Developed advanced search and filtering functionalities, along with administrative tools to generate detailed sales reports, enhancing user experience by 25% and system management

Cafe Bliss App | Java, JavaFX, Android Studio

- Developed an MVC-based Android application in Java with interactive menu browsing, dynamic cart and order management, multi-screen navigation, and RecyclerView/Singleton-based architecture for scalable data handling

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, C/C++, C#, R, HTML/CSS

Frameworks: React, Node.js, Express.js, Next.js, FastAPI, Spring Boot, Django, Flask, .NET, JUnit, JQuery

AI/ML: TensorFlow, PyTorch, Keras, Scikit-Learn, LangChain, Hugging Face, OpenAI Whisper

Cloud & DevOps: AWS, GCP, Azure, Docker/Kubernetes, GitHub Actions

Databases: MongoDB, PostgreSQL, MySQL, SQL Server, Neo4j, SQLite, ChromaDB

Tools: Git, Postman, REST APIs, Tableau, Power BI, VS Code, Android Studio, MATLAB, RStudio, Copilot Studio, Power Automate

Libraries: Redux, Three.js, NumPy, Pandas, matplotlib, spaCy, Tesseract.js